

Quant Guide

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I. SCOPE

This guide is for quantitative users who need to understand the pricing assumptions, calibration logic, risk measures, and research references used in the workstation.

II. PORTFOLIO SCOPE

The repo prices five instruments:

- a spot-starting fixed-versus-SOFR swap,
- a European swaption under a normal-volatility convention priced with Hull-White 1F plus Jamshidian's decomposition,
- a Bermudan swaption under a calibrated short-rate model,
- a second Bermudan swaption matching the workbook benchmark trade, and
- an SPX cliquet under a Black-Scholes-Merton process with analytic cliquet valuation.

III. RATES CURVE CONSTRUCTION

A. Inputs

The rates market state follows the workbook strip and a configurable valuation date, defaulting to 2026-03-10:

1D	1W	2W	3W	1M	2M
3M	6M	9M	1Y	2Y	...30Y

B. Bootstrap

The curve is constructed in `portfolio.py` with:

- one `ql.DepositRateHelper` for the 1D point, and
- a strip of `ql.OISRateHelper` objects for the remaining term SOFR OIS points.

The term structure is a `ql.PiecewiseLogCubicDiscount`. Calibration quality is checked by rebuilding quoted OIS swaps and asserting near-zero NPV.

IV. EUROPEAN SWAPTION MODEL

The European swaption calibrates `ql.HullWhite` to the selected ATM normal-volatility matrix pillar and prices with `ql.JamshidianSwaptionEngine`. The matrix follows the workbook surface, with short expiries from 1M onward, annual expiries through 10Y, then 12Y, 15Y, 20Y, and 25Y, against swap tenors through 30Y. The on-screen matrix uses the exact workbook axes. When the Bermudan calibration

needs a 3Y expiry, it is linearly interpolated between the workbook 2Y and 4Y rows.

V. BERMUDAN SWAPTION MODEL

A. Calibration

The Bermudan stack:

- 1) identifies the trade's Bermudan call schedule and the remaining swap tenor implied by the common fixed final maturity,
- 2) creates one `ql.SwaptionHelper` per diagonal pillar,
- 3) calibrates either `ql.HullWhite` or `ql.G2` while holding the mean-reversion input fixed, and
- 4) prices the Bermudan with `ql.TreeSwaptionEngine`.

All rates trades are modeled as `ql.OvernightIndexedSwap` instruments with daily compounded SOFR coupons over the chosen reset period. Payment frequency and reset frequency are explicit trade inputs; the workbook reference trade uses annual pay and annual reset. For example, a 5Y NC 2Y Bermudan keeps the same final maturity date and therefore exercises into the remaining 3Y, 2Y, and 1Y swaps along its annual call schedule. The workstation defaults Bermudans to Hull-White 1F, and the detail page allows a switch to G2++ so users can compare values and risk outputs under the same market data.

B. Interpretation

The calibration helpers are labeled by the booked Bermudan schedule, for example 1Y x 4Y, 2Y x 3Y, 3Y x 2Y, 4Y x 1Y for a five-year annual Bermudan. When a booked expiry is not on the editable matrix axes, the helper linearly interpolates in the expiry dimension and keeps the exact source matrix points for audit purposes.

VI. SPX CLIQUET MODEL

A. State Variables

The cliquet reads:

- SPX spot,
- flat dividend yield,
- flat Black volatility,
- quantity,
- percentage moneyness,
- maturity in months, and
- reset frequency in months.

B. Process And Engine

The market process is a `ql.BlackScholesMertonProcess` with:

- the shared SOFR curve used as the risk-free curve,
- a flat dividend curve, and
- a flat Black volatility surface.

The instrument is a `ql.CliquetOption` with `ql.PercentageStrikePayoff`. It is priced with `ql.AnalyticCliquetEngine`.

C. Extra Analytics

The cliquet page supplements the QuantLib price with:

- reset-period decomposition into forward-start option values,
- deterministic spot-volatility scenarios,
- trade-level analytic Greeks, and
- Monte Carlo payoff-distribution diagnostics.

VII. RISK OUTPUTS

A. Rates Trades

The swap and swaptions expose on-demand point sensitivities:

- curve ladder from single-point SOFR quote bumps,
- European swaption vega from a single matrix-cell bump,
- Bermudan matrix sensitivity from the trade schedule's helper source points, and
- a separate short-rate mean-reversion sensitivity for European and Bermudan swaptions.

B. Cliquet Trade

The cliquet page reports:

- delta, gamma, vega, theta, rho, and dividend rho,
- scenario NPVs for spot and volatility shocks, and
- distribution metrics such as 5th percentile PV and expected shortfall.

VIII. VALIDATION

The test suite validates the pricing stack in three distinct ways:

- calibration repricing for the SOFR OIS curve,
- calibration repricing for Bermudan helpers implied by the fixed-maturity call schedule, and
- ten cliquet identity cases where the payoff reduces to simpler known structures.

IX. RESEARCH REFERENCES

The curated references live in `docs/RESEARCH.md` and on the in-app model page. The swaption list emphasizes:

- Peter Caspers' QuantLib-linked swaption implementation notes,
- Jamshidian's bond-option decomposition, and
- classic Bermudan exercise papers.

The cliquet list covers numerical methods, hedging, and Levy models. It also includes regime-switching and counterparty-risk extensions.

X. MODEL BOUNDARIES

This workstation deliberately avoids:

- live data feeds,
- smile or skew modeling for the cliquet,
- short-rate models beyond the implemented Hull-White 1F and G2++ calibration paths,
- database-backed trade lifecycle, and
- portfolio aggregation beyond the five-trade sandbox.